

Digital Image Processing: Sample Questions

1. Define an image. Classify imaging techniques based on electromagnetic spectrum of light and define each technique with its application areas. Describe the basic steps of digital image processing. With block diagram, describe the approaches and components of image processing.
2. Draw the cross-section of human eye. Describe the functions of the main components of human eye pertaining to visual system.
3. With neat diagram, describe the image formation by HVS. With neat sketches, describe briefly brightness adaptation, discrimination and illusion.
4. Define image sensor. With neat sketches, describe image acquisition by different arrangements of sensors.
5. With neat sketches, briefly describe Neighborhood, Adjacency, Connectivity, Paths, Regions and Boundaries of an image pixel.
6. Define image enhancement and write its purpose. Describe intensity transformation using log and gamma transformations and discuss the effects of changing their parameters.
+ **Math**

7. Define histogram equalization and explain its effects. Show that transformation of intensity histogram results in a uniform PDF. The intensity values of an image have the PDF as:

Equation of $p(r)$

Find the transformation function for equalizing the image histogram.

8. Explain histogram matching. The intensity values of an image have the PDF as:
Equation of $p(r)$

Find the transformation function that will produce an image whose intensity PDF is

Equation of $p(z)$

9. A 3-bit image of size $M \times N$ pixels has the intensity distribution shown in following table. Get the histogram equalization transformation function and draw the original and equalized histograms.

r_k								
n_k								
z_q								
$p_z(z_q)$								

If a specified histogram of the above problem is given by the 3rd and 4th rows in the above table, find the histogram transformation function and make the output image with the specified histogram.

10. Define 2-D Fourier transform. With an example, describe the correspondence of filtering in spatial and frequency domains. Describe homomorphic filter. + **Math**

11. Describe the characteristics of 2-D ideal, Butterworth and Gaussian LPF both in frequency and spatial domains. Compare the performance of the filters for image enhancement.
12. Write the sources of image noise. Describe the characteristics of noise normally present in digital image. Briefly describe the effects of different noise on image.
13. Describe how parameters of image noise can be estimated. Describe different spatial filters used for image restoration and compare their performance. + **Math**
14. With neat diagrams, describe the image reconstruction using CT.
15. Define image segmentation and write its uses. Explain line and edge detections. Write the basic algorithms of Canny edge detector.
16. Explain local and regional processing and write their steps of operation.
17. Define erosion, dilation, opening and closing of an image and describe their effects. Briefly describe the basic algorithms of morphological operation. Explain gray-scale morphology. + **Math**
18. Describe the absorption of light by red, green and blue cones in human eye. Draw the conceptual relationship between RGB and HIS models and describe the conversion of RGB to HIS and vice versa.
19. Define data compression and write its necessity. Describe different coding schemes. Write the steps of Huffman coding. + **Math**
20. Define fidelity criteria of image compression. Draw a model of image compression. Briefly describe JPEG, JPEG 2000, PDF and BMP standards for image compression.